A Vertically Resolved, Lifecycle-Based Diagnostic Framework: A Data-Driven Approach to Monsoon Rainfall Processes

Moufeng Wan[1], Hui Su[1], and Pak Wai Chan[2]

1 Department of Civil and Environmental Engineering, The Hong Kong University of Science and Technology, Clear Water Bay, Hong Kong, China

2 Hong Kong Observatory, Kowloon, Hong Kong, China

Corresponding author: Hui Su (cehsu@ust.hk)

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1 Introduction and objectives

Monsoon rainfall prediction remains challenging, especially for heavy and extreme events that cause flooding. Traditional column-integrated moisture metrics (column water vapor or column relative humidity) obscure the vertical humidity structure that governs whether convection remains shallow or develops into deep and organized. Mid-tropospheric dryness can suppress deep convection regardless of boundary-layer moisture (Derbyshire et al., 2004; Holloway & Neelin, 2009), yet most operational frameworks do not resolve this vertical control. The importance of vertical moisture structure for convection has been further quantified through entrainment-based frameworks (Ahmed & Neelin, 2018; Schiro & Neelin, 2019). In this study, we introduce a vertically resolved, lifecycle-based diagnostic framework that tracks the evolving moisture environment across the precipitation lifecycle. The core contributions are: (i) observational evidence of state-space contraction preceding extreme rainfall; and (ii) a lifecycle-based framework that isolates event-specific anomalies from seasonal baselines. This work aligns with Climate Informatics themes in prediction and nowcasting, data-driven diagnostic methods, and reproducible, statistically rigorous evaluation.

2 Methods and data

The analysis uses 19 years (2005–2023) of hourly rain gauge data from 45 automatic weather stations and ERA5 reanalysis at 0.25° horizontal resolution on 37 pressure levels. The precipitation lifecycle framework comprises five stages: No Rain (NR)—quiescent baseline (≥12 h from any event); Before Rain (BR12–BR1)—12 h preceding onset; During Rain (DR) — active rain; After Rain (AR1–AR12)—12 h after cessation. Rainfall intensity is classified by grid-averaged maximum hourly rainfall rate during DR into five seasonal-percentile categories: R1 (drizzle, <50th), R2 (light, 50th–70th), R3 (moderate, 70th–90th), R4 (heavy, 90th–95th), and R5 (extreme, ≥95th). Statistical separability is assessed with Kolmogorov-Smirnov (KS) and Anderson-Darling (AD) tests, and Cohen's d effect size for practical significance. Nonparametric bootstrap resampling (1500 iterations) provides 95% confidence intervals. Integrated Relative Humidity (IRH) is defined as the cumulative saturation fraction from the surface *Ps* to a specified cut-off pressure *Pc*:

$$IRH(P_c) = \frac{\int_{P_c}^{P_s} q dP}{\int_{P_c}^{P_s} q_s dP}$$

where q is specific humidity and q_s is saturation specific humidity. IRH physically represents the total entrainment-related dilution a rising parcel would encounter. IRH is computed for eight pressure cut-offs from 1000–925 hPa to 1000–150 hPa, denoted as IRH_{925} to IRH_{150}.

3 Results

State-space contraction preceding extreme rainfall: For heavy and extreme rainfall (R4–R5), the interquartile range of IRH (Fig. 1) collapses by 40–76% during DR, indicating

environmental homogenization. Light rain shows no contraction. This is the regional-scale observational evidence of state-space contraction preceding extreme monsoon rainfall, consistent with convective self-aggregation theory (Wing et al., 2017). Seasonal invariance and operational thresholds: Baseline IRH differs by up to 24% across seasons (e.g., winter vs. summer), but anomaly-based thresholds (Fig.1 lines) converge: ΔIRH_{500} at DR varies with a coefficient of variation <10.4%. Extreme events require near-full-column saturation ($IRH_{500} \geq 0.86$) across all seasons, with Cohen's d effect sizes up to 1.0 for IRH_{500} between moderate-to-heavy and heavy-to-extreme rainfall events, enabling a single year-round operational criterion.

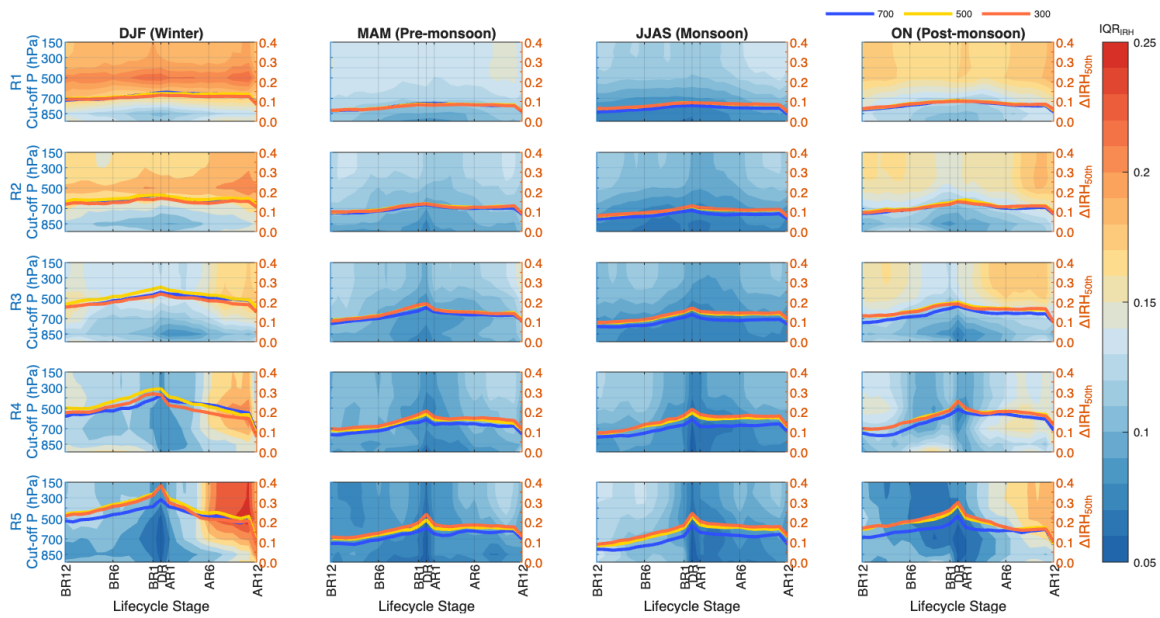

Figure 1. Lifecycle interquartile range (IQR, color shading) of IRH_{500} and IRH anomaly (ΔIRH, lines) relative to the seasonal no-rain baseline across four monsoon seasons (DJF, MAM, JJAS, ON).

The framework provides a physically grounded, vertically resolved diagnostic for operational nowcasting: monitoring IRH_{500} offers a robust indication of lifecycle stage and eventual intensity. Seasonal invariance allows for a single threshold to be set year-round. Limitations: Single-site analysis (Hong Kong); ERA5 dry biases above 500 hPa; 0.25° resolution cannot resolve individual convective cells. Future work will extend to other monsoon regions and integrate higher-resolution models.

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